

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)

Activity Tool : Short Answer Text(High)
Assessment Tool Weightage: 30%

Dr

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	UPS Product Tracking Information System UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute. Create an Entity Relationship diagram that captures this	BL1 – Remember	5

	information about the UPS system.		
2	Course Management System Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as • course with attributes name, department, and c-number • section with attributes s-number and enrollment, and dependent as a weak entity set on course • room with attributes r-number, capacity, and building Show an E-R diagram illustrating the use of all three additional entity sets listed.	BL2 – Understand	5
3	University Registrar's Office Management System A university registrar's office maintains data about the following entities: (a) courses, including number, title, credits, syllabus, and prerequisites; (b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; (c) students, including student-id, name, and program; (d) instructors, including identification number, name, department, and title. (e) Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office.	BL2 – Understand	5
4	Hospital Management System Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests	BL2 – Understand	5

	and examinations conducted. Construct appropriate tables for the above ER Diagram : Patient(SS#, name, insurance) Physician (name, specialization) Test-log(SS#, test-name, date, time) Doctor-patient (physician-name, SS#) Patient-history(SS#, test-name, date)		
5	Company Project Management System Construct an ER Diagram for Company having following details : Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations. A department controls a number of PROJECT. Projects have a unique name, number and a single location. Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee. Employee's DEPENDENT are tracked for health insurance purposes (dependentname, birthdate, relationship to employee).	BL2 – Understand	5
6	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember	5
7	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand	5
8	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand	5
9	Design an EER diagram for a Hospital Management	BL2 –	5

	System incorporating inheritance and weak entities.	Understand	
10	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember	5

Code of Conduct:

I D . S A N J A N A (1 9 2 4 7 1 0 2 9) _ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-	Understandable with minor	Somewhat unclear or	Difficult to understand

		organized answer	issues	poorly structured	
--	--	---------------------	--------	----------------------	--

ANSWER:

1. Problem Statement

UPS maintains a company-wide product tracking system to monitor the processing and current location of every shipped item. Each shipment is received at a retail center and transported through one or more transportation events before reaching its final destination. The objective is to design an **Entity Relationship (ER) Diagram** to represent this information.

Entities and Attributes

1. Shipped_Item

This is the main entity that stores information about every shipment.

Attributes

- **ItemNumber (Primary Key)**
- Weight
- Dimensions
- InsuranceAmount
- Destination
- FinalDeliveryDate

2. Retail_Center

Represents the UPS retail centers where shipments are initially received.

Attributes

- **CenterID (Primary Key)**
- CenterType
- Address

3. Transportation_Event

Represents the transportation process used to move shipments.

Attributes

- **ScheduleNumber (Primary Key)**
- EventType (Flight/Truck)
- DeliveryRoute

Relationships

1. Receives

- A **Retail Center** receives shipped items.
- One retail center can receive many shipped items.
- Each shipped item is received at only one retail center.

Relationship: Receives

Cardinality: 1 : M

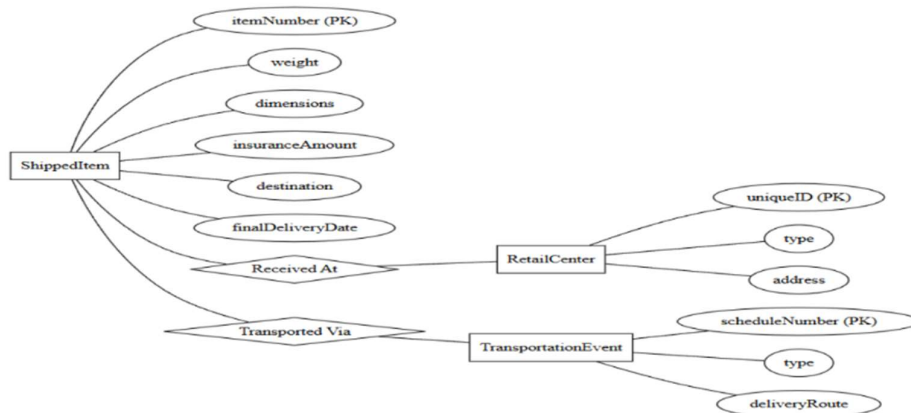
2. Transported_By

- A shipped item may travel through several transportation events.
- A transportation event can carry many shipped items.

Relationship: Transported_By

Cardinality: M : N

Since this is a many-to-many relationship, it can be implemented using an associative entity named **Shipment_Transport**.



2.ANSWER:

2. Course Management System

Problem Statement

A university maintains a database to schedule classrooms for final examinations. The system stores details about courses, their sections, examination schedules, and examination rooms. Each course may have multiple sections, and every section has a separate final examination. The objective is to design an **Entity Relationship (ER) Diagram** that represents the course, section, examination, and room details while maintaining the relationships among them.

Entities and Attributes

1. Course

The **Course** entity stores information about all the courses offered by the university.

Attributes

- **Course_Name (Primary Key)** – Unique name of the course.
- **Department** – Department offering the course.
- **Course_Number** – Unique identification number assigned to the course.

2. Section (Weak Entity)

The **Section** entity represents different sections of a course. Since a section cannot exist without a corresponding course, it is considered a **weak entity**.

Attributes

- **Section_Number (Partial Key)** – Identifies a section within a course.
- **Enrollment** – Number of students enrolled in the section.

3. Exam

The **Exam** entity stores information about the final examination conducted for each section.

Attributes

- **Exam_ID (Primary Key)** (*Optional*)
- **Time** – Scheduled time for the examination.

4. Room

The **Room** entity stores information about classrooms or examination halls used to conduct exams.

Attributes

- **Room_Number (Primary Key)** – Unique room number.
- **Capacity** – Maximum number of students the room can accommodate.
- **Building** – Name of the building in which the room is located.

Relationships

1. Has

- A **Course** can have multiple **Sections**.
- Each **Section** belongs to only one **Course**.
- Since a section depends on the course for its existence, it is modeled as a **weak entity**.

Relationship: Has

Cardinality: 1 : M

2. Has_Exam

- Every **Section** has one final **Exam**.
- Each **Exam** is associated with only one section.

Relationship: Has_Exam

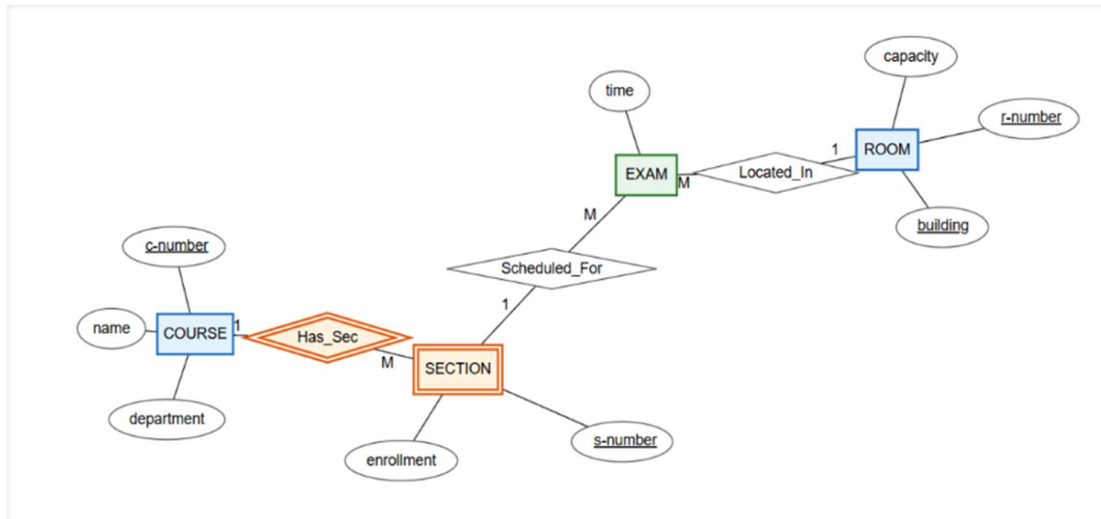
Cardinality: 1 : 1

3. Conducted_In

- Every **Exam** is conducted in one **Room**.
- A **Room** can be used for multiple examinations at different time slots.

Relationship: Conducted_In

Cardinality: M : 1



3.ANSWER:

3. University Registrar's Office Management System

Problem Statement

A university registrar's office maintains information about **courses, course offerings, students, instructors, enrollments, and grades**. The system is responsible for managing course registration, faculty assignments, classroom scheduling, student enrollment, and grade records. Each course may be offered in different semesters, students can enroll in multiple courses, and instructors teach various course offerings. The objective is to design an **Entity Relationship (ER) Diagram** that represents these entities and their relationships to efficiently manage the university's academic information.

Entities and Attributes

1. Course

The **Course** entity stores information about all courses offered by the university.

Attributes

- **Course_Number (Primary Key)** – Unique identification number of the course.
- **Title** – Name of the course.
- **Credits** – Number of credit hours assigned to the course.
- **Syllabus** – Course syllabus or curriculum.
- **Prerequisites** – Courses that must be completed before taking the course.

2. Course_Offering

The **Course_Offering** entity represents a specific offering of a course during a semester and academic year.

Attributes

- **Offering_ID (Primary Key)** – Unique identifier for each course offering.
- **Year** – Academic year in which the course is offered.
- **Semester** – Semester (Odd/Even or Spring/Fall).
- **Section_Number** – Section assigned to the course offering.
- **Timings** – Class schedule.

- **Classroom** – Room where the class is conducted.

3. Student

The **Student** entity stores details of students enrolled in the university.

Attributes

- **Student_ID (Primary Key)** – Unique identification number of the student.
- **Name** – Student's name.
- **Program** – Degree program or department.

4. Instructor

The **Instructor** entity stores information about faculty members.

Attributes

- **Instructor_ID (Primary Key)** – Unique identification number of the instructor.
- **Name** – Instructor's name.
- **Department** – Department to which the instructor belongs.
- **Title** – Designation (Professor, Assistant Professor, etc.).

5. Enrollment (Associative Entity)

The **Enrollment** entity records the enrollment of students in course offerings and stores the grade obtained by each student.

Attributes

- **Student_ID (Foreign Key)** – References the Student entity.
- **Offering_ID (Foreign Key)** – References the Course_Offering entity.
- **Grade** – Grade awarded to the student.

Relationships

1. Offers

- A **Course** can have **multiple Course Offerings** in different semesters and academic years.
- Each **Course Offering** belongs to only **one Course**.

Relationship: Offers

Cardinality: 1 : M (One-to-Many)

2. Teaches

- An **Instructor** can teach **one or more Course Offerings**.
- Each **Course Offering** is assigned to **one Instructor**.

Relationship: Teaches

Cardinality: 1 : M (One-to-Many)

3. Enrolls

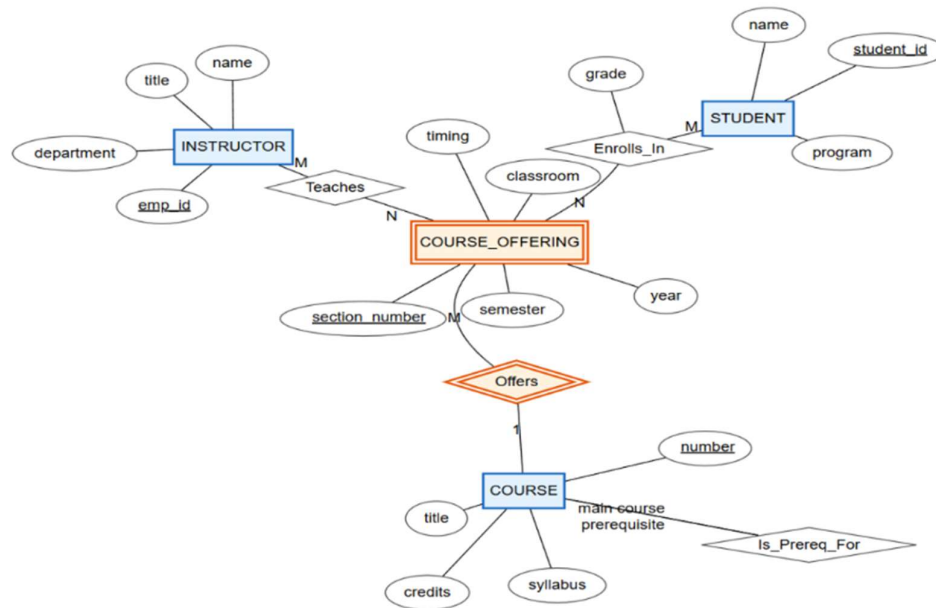
- A **Student** can enroll in **multiple Course Offerings**.
- Each **Course Offering** can have **many Students** enrolled.

Relationship: Enrolls

Cardinality: M : N (Many-to-Many)

Since this is a many-to-many relationship, it is implemented using the associative

entity **Enrollment**, which also stores the **Grade** obtained by the student.



4.ANSWER:

4. Hospital Management System

Problem Statement

A hospital maintains information about **patients, physicians, medical tests, and patient history**. Each patient may undergo several medical tests and examinations, while physicians diagnose and treat multiple patients. The hospital also keeps a record of all tests performed and maintains the medical history of every patient. The objective is to design an **Entity Relationship (ER) Diagram** that represents these entities, their attributes, and relationships for efficient hospital management.

Entities and Attributes

1. Patient

The **Patient** entity stores personal and insurance details of patients admitted to the hospital.

Attributes

- **SS# (Primary Key)** – Unique Social Security Number or Patient ID.
- **Name** – Name of the patient.
- **Insurance** – Insurance provider or policy details.

2. Physician

The **Physician** entity stores information about doctors working in the hospital.

Attributes

- **Physician_Name (Primary Key)** – Name or unique ID of the physician.
- **Specialization** – Medical specialty (e.g., Cardiology, Neurology, Orthopedics).

3. Test_Log

The **Test_Log** entity records all medical tests and examinations conducted for

patients.

Attributes

- **Test_ID (Primary Key)** (*Optional*)
- **SS# (Foreign Key)** – References the Patient entity.
- **Test_Name** – Name of the medical test.
- **Date** – Date on which the test was conducted.
- **Time** – Time of the test.

4. Patient_History

The **Patient_History** entity stores previous medical records and test history of each patient.

Attributes

- **History_ID (Primary Key)** (*Optional*)
- **SS# (Foreign Key)** – References the Patient entity.
- **Test_Name**
- **Date**

Relationships

1. Treats

- A **Physician** can treat **many Patients**.
- A **Patient** may be treated by **one or more Physicians**.

Relationship: Treats

Cardinality: M : N (**Many-to-Many**)

Since this is a many-to-many relationship, it can be implemented using an associative entity named **Doctor_Patient**.

2. Has_Test

- A **Patient** can undergo **multiple Medical Tests**.
- Each **Test_Log** record belongs to only **one Patient**.

Relationship: Has_Test

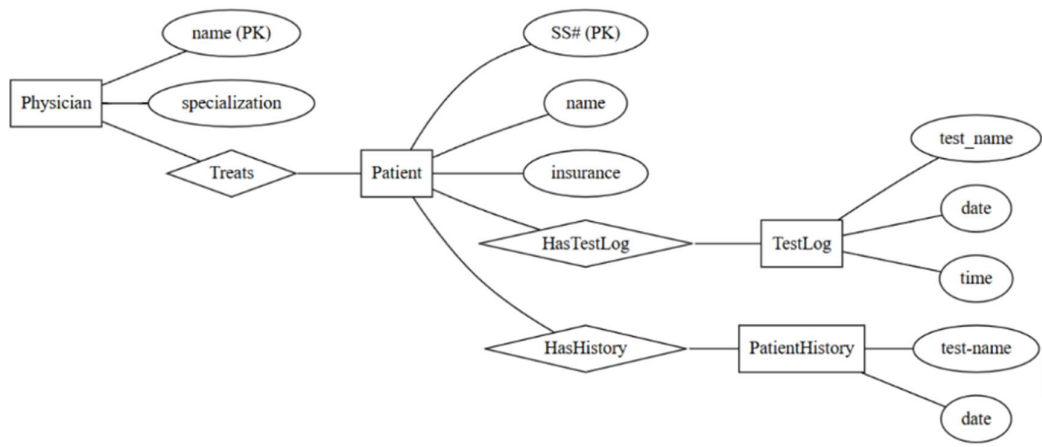
Cardinality: 1 : M (**One-to-Many**)

3. Maintains

- A **Patient** has one **Patient History** that stores records of previous tests and examinations.
- A **Patient History** belongs to only one patient.

Relationship: Maintains

Cardinality: 1 : M (**One-to-Many**)



5.ANSWER:

5. Company Project Management System

Problem Statement

A company is organized into different **departments**, each managed by an employee. Every department controls several **projects** and may operate from multiple locations. Employees work in one department but can participate in multiple projects. The company also maintains information about employees, their supervisors, and their dependents for health insurance purposes. The objective is to design an **Entity Relationship (ER) Diagram** that represents the company's organizational structure and project management system.

Entities and Attributes

1. Department

The **Department** entity stores information about each department in the company.

Attributes

- **Department_ID (Primary Key)** – Unique identifier for the department.
- **Department_Name** – Name of the department.

2. Employee

The **Employee** entity stores personal and employment details of employees.

Attributes

- **SSN (Primary Key)** – Unique employee identification number.
- **Name** – Employee's name.
- **Address** – Residential address.
- **Salary** – Employee's salary.
- **Sex** – Gender of the employee.
- **Birth_Date** – Date of birth.

3. Project

The **Project** entity stores information about projects handled by the company.

Attributes

- **Project_Number (Primary Key)** – Unique project number.
- **Project_Name** – Name of the project.
- **Location** – Location where the project is carried out.

4. Department_Location

The **Department_Location** entity stores the locations where a department operates.

Attributes

- **Location_ID (Primary Key)** (*Optional*)
- **Department_ID (Foreign Key)**
- **Location**

5. Dependent (Weak Entity)

The **Dependent** entity stores details of employees' dependents for health insurance purposes. It is a **weak entity** because it depends on the Employee entity.

Attributes

- **Dependent_Name (Partial Key)**
- **Birth_Date**
- **Relationship**

6. Works_On (Associative Entity)

The **Works_On** entity records the projects assigned to employees and the number of hours they work on each project.

Attributes

- **SSN (Foreign Key)**
- **Project_Number (Foreign Key)**
- **Hours_Per_Week**

Relationships

1. Manages

- Each **Department** is managed by **one Employee**.
- An **Employee** manages only **one Department**.

Relationship: Manages

Cardinality: 1 : 1 (**One-to-One**)

2. Controls

- A **Department** can control **multiple Projects**.
- Each **Project** belongs to only **one Department**.

Relationship: Controls

Cardinality: 1 : M (**One-to-Many**)

3. Assigned_To

- An **Employee** works in **one Department**.
- A **Department** can have **many Employees**.

Relationship: Assigned_To

Cardinality: 1 : M (One-to-Many)

4. Works_On

- An **Employee** can work on **multiple Projects**.
- A **Project** can have **multiple Employees**.

Relationship: Works_On

Cardinality: M : N (Many-to-Many)

Since this is a many-to-many relationship, it is implemented using the associative entity **Works_On**, which stores the **Hours_Per_Week** worked by each employee.

5. Has_Dependent

- An **Employee** can have **multiple Dependents**.
- Each **Dependent** belongs to only **one Employee**.

Relationship: Has_Dependent

Cardinality: 1 : M (One-to-Many)

6. Supervises

- One **Employee** can supervise **many Employees**.
- Each **Employee** reports to only **one immediate Supervisor**.

Relationship: Supervises

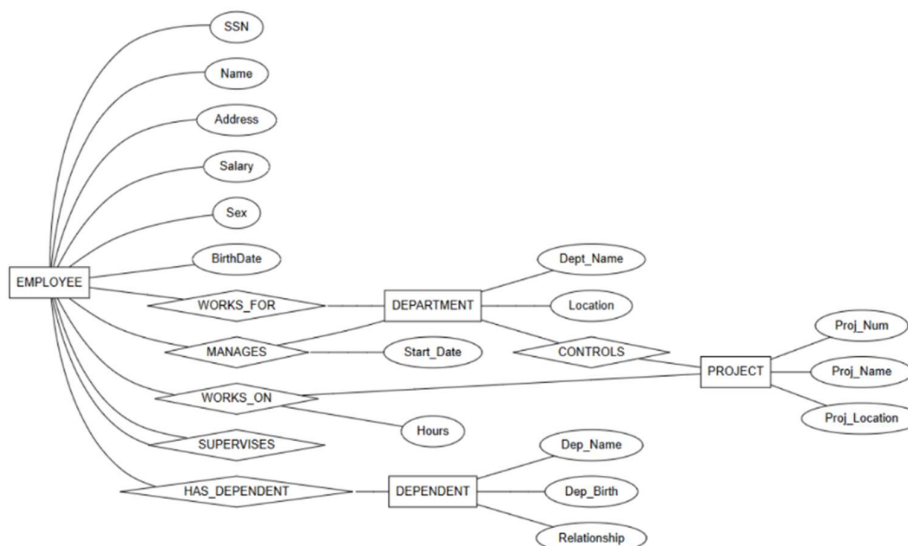
Cardinality: 1 : M (Recursive Relationship)

7. Located_At

- A **Department** may operate from **multiple Locations**.
- Each **Department Location** belongs to one Department.

Relationship: Located_At

Cardinality: 1 : M (One-to-Many)



6.ANSWER:

Explain Schema and Instance with Examples. Distinguish between Logical Schema, Physical Schema, and View Schema.

A **Schema** is the overall structure or blueprint of a database. It defines how data is organized, including the tables, attributes, relationships, constraints, and data types. The schema is designed before data is entered into the database and changes very rarely.

Example of Schema

STUDENT

Student_ID (Primary Key)

Name

Department

Age

Phone_Number

This structure represents the design of the **Student** table.

Instance

An **Instance** is the actual data stored in the database at a particular point in time. Unlike the schema, the instance changes frequently as records are inserted, updated, or deleted.

Example of Instance

Student_ID Name Department Age

101 Rahul CSE 20

102 Priya ECE 19

103 Arun IT 21

The above table is an instance of the **Student** schema.

Difference Between Schema and Instance

Schema

Schema is the structure or blueprint of the database.

It is created during database design.

Changes occur very rarely.

Also called **Intension**.

Instance

Instance is the actual data stored in the database.

It changes whenever data is inserted, updated, or deleted.

Changes occur frequently.

Also called **Extension**.

Types of Schema

1. Logical Schema

The **Logical Schema** describes the logical structure of the entire database. It defines tables, attributes, relationships, primary keys, and foreign keys without considering how the data is physically stored.

Example:

- Student(Student_ID, Name, Department)
- Course(Course_ID, Course_Name)

2. Physical Schema

The **Physical Schema** describes how the database is actually stored in the computer's memory. It includes storage structures, file organization, indexing, and access methods. This schema is managed by the Database Management System (DBMS).

Example:

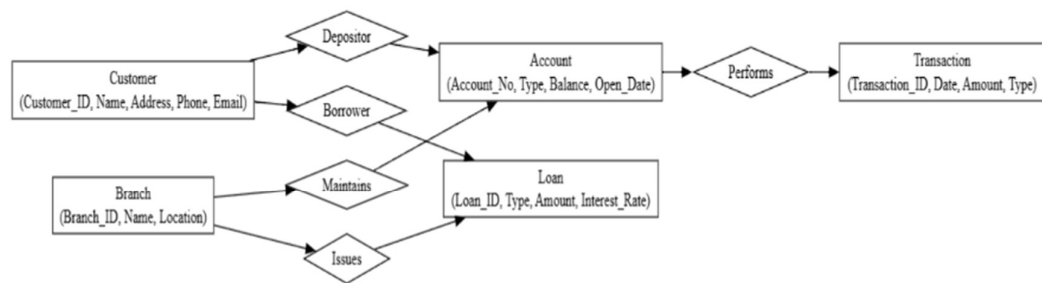
- Data stored in files on a hard disk or SSD.
- B+ Tree indexes used for faster searching.
- Hashing techniques used for quick data retrieval.

3. View Schema

The **View Schema** is a customized view of the database for different users. It displays only the required data while hiding unnecessary or sensitive information.

Example:

- A **Student** can view only their marks and personal details.
- A **Faculty Member** can view student attendance and grades.
- An **Administrator** can access all tables in the database.



7.ANSWER:

Bank Management System – ER Diagram

Problem Statement

A **Bank Management System** is used to manage the information related to customers, accounts, loans, branches, and employees. The system keeps track of customer accounts, loan details, branch information, and employees working in each branch. It also records the relationships between customers, accounts, loans, and branches. The objective is to design an **Entity Relationship (ER) Diagram** that represents these entities and their relationships for efficient banking operations.

Entities and Attributes

1. Customer

The **Customer** entity stores personal information about bank customers.

Attributes

- **Customer_ID (Primary Key)** – Unique identification number of the customer.
- **Name** – Customer's name.
- **Address** – Residential address.

- **Phone_Number** – Contact number.

2. Account

The **Account** entity stores information about bank accounts maintained by customers.

Attributes

- **Account_Number (Primary Key)** – Unique account number.
- **Account_Type** – Savings or Current account.
- **Balance** – Current account balance.

3. Loan

The **Loan** entity stores information about loans provided by the bank.

Attributes

- **Loan_Number (Primary Key)** – Unique loan number.
- **Loan_Type** – Home Loan, Personal Loan, Education Loan, etc.
- **Loan_Amount** – Amount sanctioned.
- **Interest_Rate** – Interest charged on the loan.

4. Branch

The **Branch** entity stores information about different bank branches.

Attributes

- **Branch_ID (Primary Key)** – Unique branch identification number.
- **Branch_Name** – Name of the branch.
- **Location** – Address or city where the branch is located.

5. Employee

The **Employee** entity stores details of employees working in the bank.

Attributes

- **Employee_ID (Primary Key)** – Unique employee identification number.
- **Name** – Employee's name.
- **Designation** – Job role or position.
- **Salary** – Employee's salary.

Relationships

1. Owns

- A **Customer** can own **one or more Accounts**.
- Each **Account** belongs to only one Customer.

Relationship: Owns

Cardinality: 1 : M (One-to-Many)

2. Takes

- A **Customer** can take **multiple Loans**.
- Each **Loan** is issued to only one Customer.

Relationship: Takes

Cardinality: 1 : M (One-to-Many)

3. Maintains

- A **Branch** maintains **multiple Accounts**.
- Each **Account** is maintained by only one Branch.

Relationship: Maintains

Cardinality: 1 : M (One-to-Many)

4. Works_In

- A **Branch** can have **many Employees**.
- Each **Employee** works in only one Branch.

Relationship: Works_In

Cardinality: 1 : M (One-to-Many)

8.ANSWER:

Explain the Extended ER (EER) Model. Describe the Concepts of Generalization, Specialization, and Aggregation. (5 Marks)

Extended ER (EER) Model

The **Extended Entity-Relationship (EER) Model** is an advanced version of the traditional ER model. It extends the basic ER model by introducing concepts such as **generalization, specialization, aggregation, and inheritance**. These concepts help represent complex real-world situations more accurately and make database design more flexible and efficient.

The EER model is widely used in large database systems where entities have hierarchical relationships and shared characteristics.

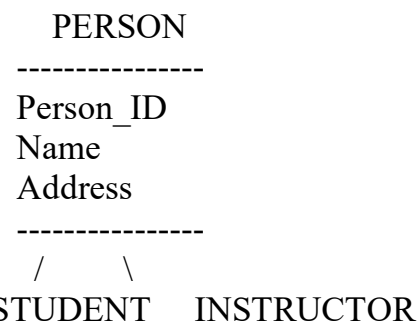
Concepts of EER Model

1. Generalization

Generalization is the process of combining two or more lower-level entities that have common features into a single higher-level entity called a **superclass**. It helps reduce redundancy by grouping common attributes.

Example

Both **Student** and **Instructor** have common attributes such as **ID, Name, and Address**. These common attributes can be placed in a superclass called **Person**.



Advantages

- Reduces duplication of common attributes.
- Simplifies database design.
- Improves data organization.

2. Specialization

Specialization is the opposite of generalization. It divides a higher-level entity (superclass) into multiple lower-level entities (subclasses) based on specific characteristics.

Example

An **Employee** can be specialized into **Manager**, **Engineer**, and **Accountant**, each having its own specific attributes.

Advantages

- Represents different types of entities clearly.
- Stores specialized information separately.
- Improves flexibility in database design.

3. Aggregation

Aggregation is a concept used when a relationship between two entities needs to be treated as a higher-level entity so that it can participate in another relationship.

Example

An **Employee** works on a **Project**. This **Works_On** relationship can be treated as a single entity called **Assignment**, which can then be supervised by a **Manager**.

EMPLOYEE ----- WORKS_ON ----- PROJECT



Advantages

- Represents complex relationships effectively.
- Simplifies modeling of real-world scenarios.
- Improves clarity of the database structure.

9.ANSWER:

Design an EER Diagram for a Hospital Management System Incorporating Inheritance and Weak Entities

Problem Statement

A **Hospital Management System** stores information about patients, doctors, nurses, and medical records. All hospital staff and patients share common personal details, while doctors and nurses have additional specialized information. The hospital also maintains medical records for each patient. Since a medical record cannot exist without a patient, it is modeled as a **weak entity**. The objective is to design an **Extended Entity-Relationship (EER) Diagram** using **inheritance (generalization/specialization)** and **weak entities**.

Entities and Attributes

1. Person (Superclass)

The **Person** entity contains common information shared by patients, doctors, and nurses.

Attributes

- **Person_ID (Primary Key)** – Unique identification number.

- **Name** – Name of the person.
- **Age** – Age of the person.
- **Gender** – Gender.
- **Address** – Residential address.
- **Phone_Number** – Contact number.

2. Patient (Subclass)

The **Patient** entity inherits the common attributes of **Person** and stores patient-specific details.

Attributes

- **Patient_ID (Primary Key)**
- **Insurance_Details**
- **Blood_Group**

3. Doctor (Subclass)

The **Doctor** entity inherits from **Person** and stores doctor-specific information.

Attributes

- **Doctor_ID (Primary Key)**
- **Specialization**
- **Experience**

4. Nurse (Subclass)

The **Nurse** entity inherits from **Person** and stores nurse-specific information.

Attributes

- **Nurse_ID (Primary Key)**
- **Shift**
- **Department**

5. Medical_Record (Weak Entity)

The **Medical_Record** entity stores the patient's diagnosis and treatment details. It is a **weak entity** because it cannot exist without a patient.

Attributes

- **Record_Number (Partial Key)**
- **Diagnosis**
- **Treatment**
- **Admission_Date**
- **Discharge_Date**

Relationships

1. Inheritance (Generalization/Specialization)

- **Patient, Doctor, and Nurse** inherit common attributes from the **Person** entity.
- This relationship represents **Generalization/Specialization (Inheritance)**.

2. Has_Record

- A **Patient** can have **multiple Medical Records**.
- Each **Medical Record** belongs to only **one Patient**.

Relationship: Has_Record

Cardinality: 1 : M (One-to-Many)

Since **Medical_Record** depends on the **Patient** entity for its existence, it is modeled as a **weak entity**.

3. Treats

- A **Doctor** can treat **many Patients**.
- A **Patient** may receive treatment from **multiple Doctors**.

Relationship: Treats

Cardinality: M : N (Many-to-Many)



10.ANSWER:

10. Explain the Role of a DBA (Database Administrator) and the Responsibilities Associated with the Position.

Database Administrator (DBA)

A **Database Administrator (DBA)** is a person responsible for managing, maintaining, securing, and monitoring a database system. The DBA ensures that the database operates efficiently, remains secure, and is always available for users and applications. A DBA also performs regular maintenance, backup, and recovery operations to protect organizational data from loss or failure.

Responsibilities of a DBA

1. Database Installation and Configuration

The DBA installs the Database Management System (DBMS), configures the database environment, and ensures that it is ready for use.

2. Database Security

The DBA provides security by creating user accounts, assigning roles and permissions, and preventing unauthorized access to the database.

3. Backup and Recovery

The DBA performs regular database backups and restores data in case of hardware failure, software failure, or accidental data loss.

4. Performance Monitoring and Tuning

The DBA continuously monitors database performance and optimizes queries, indexes, and storage to improve efficiency and response time.

5. Database Maintenance

The DBA updates the database, removes unnecessary data, manages storage space, and performs routine maintenance to ensure smooth operation.

6. User Management

The DBA creates, modifies, and deletes user accounts, manages passwords, and controls user privileges according to organizational requirements.

7. Data Integrity

The DBA ensures that the data stored in the database is accurate, consistent, and reliable by enforcing integrity constraints and validation rules.

8. Disaster Recovery

The DBA prepares disaster recovery plans to restore database operations quickly in case of system crashes, natural disasters, or cyberattacks.

Roles of a DBA

- Installs and configures database software.
- Maintains database security and access control.
- Performs regular backup and recovery operations.
- Monitors and improves database performance.
- Ensures data integrity and consistency.
- Manages users and database permissions.
- Troubleshoots database-related issues.
- Maintains high availability and reliability of the database.

Advantages of a DBA

- Ensures data security and confidentiality.
- Prevents data loss through regular backups.
- Improves database performance and efficiency.
- Maintains data consistency and integrity.
- Supports smooth and uninterrupted business operations.
- Reduces system downtime through proper maintenance and recovery planning.